

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Semester Examination – 2022

Course: B. Tech.

Branch : AI&DS

Semester: III

Subject Code & Name: Digital Logic & Signal Processing (BTESC305)

Max Marks: 60

Date:17/03/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find 2's Complement of the number i) 01001110 ii) 00110101 iii) 11011000.	CO1	
B)	Multiply i) 1001 by 1101 ii) Divide 1110101 by 1001 iii) convert 675.675 in to hexadecimal	CO1	
C)	Draw and Explain gates i) NAND ii) NOR iii) EX-OR	CO1	
Q.2	Solve Any Two of the following.		12
A)	Define the term Signal. Explain Elementary Signals used for testing with its diagram,	CO2	
B)	Sketch following signals x(n) is given as x(n) = {1,2,3,4,5} i) x(3n-1) ii) x(-n+1)	CO2	
C)	Explain Classification of System with its examples in detail.	CO2	
Q. 3	Solve Any Two of the following.		12
A)	Perform Linear Convolution of the following two sequences using matrix method. x(n) = {1,2,1,-1} h(n) = {1,2,3,0}	CO3	
B)	Find DTFT of the following signals. i) x(n) = {1,2,3,4} ii) x(n) = {0,1,2,3}	CO3	
C)	Compare DIT-FFT and DIF-FFT algorithm.	CO3	
Q.4	Solve Any Two of the following.		12
A)	Find Z transform of the following and indicate the ROC. i) x(n) = {1, 2, 3, 4} ii) x(n) = a^n u(n), a > 1	CO4	
B)	Illustrate ROC and explain its properties in detail	CO4	
C)	Find the inverse z-transform of the following using partial fraction method . i) X(Z) = $\frac{z}{3z^2-4z+1}$ z > 1	CO4	
Q. 5	Solve Any One of the following.		12
A)	. Explain Correlation function and Properties of autocorrelation.	CO5	
B)	Explain properties of i) ESD ii) PSD.	CO5	